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LIGHTING PRACTICE: INTRODUCTION TO RESILIENT LIGHTING SYSTEMS

AN AMERICAN NATIONAL STANDARD

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should be directed to IES.

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By the Resilience Committee

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1.0 Introduction and Scope

1.1 Introduction

Every year, buildings and infrastructure are subjected to severely disruptive events caused by natural disasters and human-made intentional attacks. These events create significant disruptions and economic damages in addition to unfortunate injury and loss of life. According to the U.S. National Oceanic and Atmospheric Administration, there were 22 separate billion-dollar extreme weather events across the United States in 2020 resulting in $95 billon of damages.[1] To reduce the impacts of these disasters, communities, building owners, institutions, public agencies, and others are looking to enhance the resilience of buildings, infrastructure, and communities overall.

For the purpose of this document, *resilient design* is defined as the ability to prepare and plan for, absorb, recover from, and more successfully adapt to adverse events.[2] Emergency planning is not a new undertaking. However, with such high-profile events as Hurricane Katrina, Superstorm Sandy, and the COVID-19 pandemic, focused efforts to understand how the built environment can incorporate resiliency into homes, buildings, and infrastructure is a distinct goal. The purpose of this Lighting Practice document (LP) is to introduce the concept of resilient design and explain how lighting systems can support the goals of enhancing the resilience of buildings. The intent is to provide guidance on lighting performance, controls, and the characteristics of lighting equipment for resilient buildings.

1.2 Scope

The scope of this LP does not include emergency lighting for the purpose of evacuation and marking of egress paths, since building codes already dictate these needs. In addition, this document is not intended for the development of emergency plans or courses of action during extreme events. For example, determining when to evacuate a building or whether to shelter in place is determined by local authorities.

2.0 Definitions

2.1 Resiliency

The term *resiliency* is often used to describe the broad goal of community resilience, which is concerned with addressing functionality of services across a community. Buildings and infrastructure play critical roles in the community; therefore, it is critical that the design of resilient buildings (resilient design), take into consideration community-wide goals. On the other hand, building owners and institutions may have resiliency goals independent of community needs, which will also need to be considered by designers.

Current building codes focus on life safety and dictate how buildings should sustain acute events such as fires and earthquakes, while facilitating building evacuation. The ways that buildings anticipate, adapt, and function through sustained disruptive events may have common approaches to building evacuation, but the goals are distinct from those of evacuation. For example, there is industry consensus and, through model codes and standards such as the *International Building Code* and the U.S. *National Electric Code,* guidance for egress lighting (see **Section 3.0 The Role of Lighting in Resilient Buildings and Infrastructure**).

However, if adverse events require occupants to shelter in place, are there minimal building functions that require lighting? At what levels? For how long? Furthermore, while there are ways to “harden” systems to sustain adverse events, do all systems need to be hardened? Resilient design attempts to address these questions. The broad implication of the phrase *adverse* events can not only encompass a range of natural disasters from tornados to floods, but can also include human-made events that compromise building systems and energy service, such as terrorist attacks and cyber hacking. Planning for and adapting to these widely differing adverse events can result in different and potentially conflicting solutions. For example, planning for flood conditions typically dictates raising critical systems above flood elevations, whereas planning for tornados may encompass storm shelters that are in basements or below grade.

2.2 Other Terms

auxiliary power: (See **Sidebar** on page 6.)

base flood elevation (BFE): How high floodwater is likely to rise during a flood that has a one percent chance of occurring in any given year.[3] The BFE is determined by statistical analysis for each local area, and designated on flood insurance rate maps (FIRM). The BFE is also known as the 100-year-flood elevation.

breakaway wall: Any type of wall subject to flooding that is not required to provide structural support to a building or other structure and that is designed and constructed such that, under base flood or lesser flood conditions, it will collapse under specific lateral loads.

coastal high hazard area (CHHA): Areas where waves and fast-moving water can cause extensive damage during a base food event.[3]

continuity of operations plan (COOP): A formal plan that identifies essential organizational functions and operational procedures during disruptive events.

design flood: The flood associated with the greater of: 1) a community's flood hazard map, designated by the responsible design professional; or 2) an otherwise legally designated flood.

design flood elevation (DFE): The elevation of the *design flood*, including wave height, relative to the datum specified on a community's flood hazard map.

dry floodproofing: Techniques used to make equipment watertight, with all elements substantially impermeable.

flooding: A general condition of partial or complete inundation of normally dry land from the overflow of inland or tidal waters, or directly from heavy rainfall.

Ingress Protection: A classification system published by the International Electrotechnical Commission that rates the degree of protection provided by mechanical casings and electrical enclosures against intrusion, dust, accidental, contact, and water.

passive survivability: A building's ability to maintain selected system functions without the use of energy.

system hardening: Construction techniques designed to sustain the impacts of extreme events.

wet floodproofing: Techniques such as use of damage-resistant material for floors, used to prevent flood damage to equipment located in areas allowed to flood.

3.0 The Role of Lighting in Resilient Buildings and Infrastructure

Most building codes require that architects and engineers design emergency and egress lighting for the purpose of evacuation into nearly all building projects. State, provincial, and federal jurisdictions often enact requirements using model codes such as the International Building Code,[4] NFPA 70,[5] or NFPA 101.[6] Ultimately, these standards require minimal illumination, typically about 1 lux (0.1 fc), along egress paths for limited durations, typically 90 minutes to facilitate immediate evacuation of a building.

Evacuation may be appropriate for extreme events that create unsafe conditions within a building; however, there are instances where sheltering in place within a building is warranted. Lighting for this purpose can be characterized as lighting for critical or continued operations. NFPA 70 provides suggested requirements for distributing power to critical operations power systems (COPS), but lighting requirements needed for visibility, safety, and security are not prescribed. Critical operations may include lighting systems for basic building functions such as restrooms and kitchens, or for operations deemed necessary for emergency response. Furthermore, local or regional emergency planners may dictate selected private-industry functions as critical to community needs, such as food, fuel, and even banking, and lighting systems will need to provide functional illumination, which may be lower than generally recommended by IES, for these continued operations.

4.0 Programming for Resilient Lighting

As with any design, it is important to program and document the basis of design (BOD). While an office building, school, hospital, grocery store, or high-rise condominium may have independent goals for resilience, it is also important to understand that regional responses to extreme events may impact services affecting the ability of the building to operate, as well as the overall function of the building. Public schools may act as emergency shelters to house displaced persons. Other buildings, such as warehouses, may be converted to emergency distribution centers. Air-conditioned senior centers or YMCA buildings are sometimes used for emergency cooling centers during extreme heat. Resiliency goals for buildings are like the layers of skin on an onion (see **Figure 4-1**). They need to consider not only building-level needs, but also the larger needs and roles that the building may have in the larger community.

The larger goal of programming for resiliency to take into account local and regional considerations is beyond the scope of this document, but lighting designers

Figure 4-1. The resiliency goals for a building need to satisfy multiple layers of needs. (Graphic courtesy of Building Resilience LA)

need to understand these goals in order to determine potential impacts on the lighting system. The primary questions that need to be addressed are:

- Are there elements in the lighting design that can facilitate a building's recovery from disruptive events?
- Will the building be evacuated during disruptive events, and if so, are there needs beyond those covered by current emergency lighting codes?
- Will the building be expected to operate during disruptive events?
  - What functions will the building provide during the disruptive events? For example:
    - Sleeping
    - Activities
    - Food preparation and eating
    - Hygiene
    - Emergency operations
    - Emergency electrical distribution
  - Will selected building areas change function? For example:
    - Gymnasiums converted to sleeping areas
    - Conference rooms converted to emergency operations centers
  - How long will the building need to operate under these conditions?
  - Does lighting system equipment need to function under extreme environmental conditions? For example:
    - Water infiltration
    - Extreme heat or cold

While detailed programming of a building's resiliency may be managed by the architect or a specialized consultant, the lighting designer should be prepared to ask these and other questions to understand lighting system needs.

5.0 Sustaining Disruptive Events

It is important to emphasize that it is not the intent of this document to determine *whether* occupants should evacuate or shelter in place. Occupants should always follow mandatory evacuation instructions from local emergency managers. However, there are situations where it is critical that buildings continue to function following disruptive events, in part or completely, so that occupants can seek shelter or maintain selected building operations. Lighting plays a critical role in allowing personnel to perform building operations deemed essential, in addition to facilitating the safe, comfortable sheltering of occupants.

5.1 Lighting for Continuing Operations

5.1.1 What Needs to Be Lighted? To determine what needs to be lighted, essential building functions need to first be identified. If available, a continuity of operations plan (COOP) may explicitly identify building needs or may indirectly indicate functions that could be used to determine building needs. In the end, the design team needs to discuss end user needs and expectations for continuing operations. Do areas supporting these functions need full lighting, or may illumination levels be reduced? Will any of these areas act as temporary shelter? Are alternative lighting control schemes required? It is also important to remember that there are basic functions for building occupation, such as circulation, sanitation, security, and maintenance of equipment. Suggested lighting needs for continuing operations include:

- Areas identified as requiring continuing operations
- Restrooms, showers
- Corridors and circulation
- Food preparation and dining areas
- Exterior entries and paths of egress
- Mechanical and electrical rooms with equipment operating during a sustained event
- Storm safe rooms

5.1.2 Illumination Levels. Numerous IES Recommended Practice documents provide design recommendations for specific applications and building uses. Stressing quality lighting, these documents address lighting for human needs, including lighting levels for task performance and visual comfort, as well as architectural considerations such as how lighting renders and reinforces the composition of the built environment. For the purposes of lighting for continuing operations, the focus should be on providing sufficient lighting to perform essential functions while at the same time maintaining an environment that is comfortable, safe, and reassuring. Recommended illuminance levels from IES standards should be followed where possible, but judgment will be necessary to distinguish and prioritize functional lighting needs over aesthetic needs under extreme conditions.

The fact that people may be sheltering in place or occupying sparsely populated buildings could create environments that appear ominous or unsecure if lighting approaches do not consider security. *IES G-1-16, Security Lighting for People, Property, and Critical Infrastructure* provides valuable guidance in this concern.[7]

Lighting levels for specialized needs such as emergency operations centers or areas providing medical care may need to be higher (see, for example, ANSI/IES RP-29-20[8]).

5.1.3 Safe Rooms. Several documents from the International Code Council and the U.S. Federal Emergency Management Administration[9,10,11] provide recommendations for the construction of safe rooms. Safe rooms are designed to provide occupants protection from high winds associated with tornadoes and hurricanes. These guidance documents provide different requirements for tornadoes than for hurricanes because of the differing duration and characteristics of these events.

5.1.4 Passive Survivability. With respect to lighting, approaches to passive survivability predominantly consist of maximizing daylighting. Passive resilient design approaches may consist of locating important building functions near windows, clerestories, and skylights in order to eliminate or minimize the need for electric lighting.

LED light sources are inherently DC in nature. Therefore, approaches to provide separate or portable lighting equipment that can be powered by battery or photovoltaic means should be explored.

5.1.5 Power. Electricity is needed to power lighting equipment. When buildings and infrastructure lose utility electrical power, auxiliary systems, typically generators, provide backup for selected loads. Battery units, centralized units, and even photovoltaic systems provide alternatives to generator systems but typically are more limited in duration.

5.1.5.1 Duration. While an auxiliary power system, whether battery or generator powered, may be designed by electrical engineers, it is important for the lighting designer to understand the duration of lighting operation that will be provided. This may vary extensively depending on the end user, jurisdiction, or event type. **Table 5-1** provides some suggested durations.

5.1.5.2 Characterization of Power. The term *emergency power* is often generically used to describe auxiliary power systems, but most U.S. jurisdictions (and some parts of Mexico[12]) adopt the *National Electrical Code* (NEC),[5] which characterizes different levels of auxiliary power, depending on the importance of the loads served (see **Sidebar**, page 6). It is important to understand the distinctions between these categories because the NEC, and potentially other codes (e.g., the National Building Code of Canada[13]), provides differing restrictions and burdens on circuiting and control among the different classifications. In particular, the lighting described in this document does not have an NEC definition at this time and so falls into the category of optional standby loads, which according to the NEC cannot be connected to the emergency power portion of the auxiliary power system. Therefore, the electrical and/or life-safety code applicable to the project should be consulted.

5.1.5.3 Peripheral Equipment. While the focus of auxiliary power for lighting is on circuiting to power the luminaires, it is critical that power to lighting controls equipment not be forgotten. Line voltage equipment such as single-pole switches and occupancy sensors will need to maintain their function. Other associated equipment may also require auxiliary power to maintain lighting function. This is particularly relevant as more lighting control systems are embedded in building IT networks and incorporated into luminaires themselves. Some of the equipment that should be investigated to confirm whether auxiliary power should be provided for purposes of resiliency includes:

- Lighting circuits extending through lighting control panels that provide power
- Control panels with separate circuiting
- IT servers and routers that control lighting
- Distributed low-voltage lighting control systems
- Wireless receivers and controls
- Luminaire-level lighting controls

Table 5-1. Suggested Duration for Auxiliary Power Systems

Type of Facility	Suggested Duration
Baseline facilities: Residential, buildings, hospitals, lodging, nursing homes, emergency shelters, and emergency facilities including fire stations, police stations, 911 call centers	Four consecutive days, 24 hours per day
Fundamental community services: Pharmacies, convenience stores, grocery stores, and facilities with significant stocks of refrigerated frozen food	**General operations:** Four consecutive days, 8 hours per day, during daylight hours. **Refrigeration:** Four consecutive days, 24 hours per day
Gas stations	Four consecutive days, 12 hours per day during daylight hours for general operations
ATMs	Powered during regular business hours
Hurricane safe room	Minimum 24 hours
Tornado safe room	Minimum 2 hours

Sources: Table derived in part from LEED Credit; Safe room requirements per ICC 500[9]

SIDEBAR – NEC Characterization of Auxiliary Power Systems[5]

Emergency power:
"...those systems legally required and classed as emergency by municipal, state, federal or other codes or by governmental agency having jurisdiction. These systems are intended to automatically supply illumination or power, or both, to designated areas and equipment in the event of failure of the normal supply or in the event of accident to elements of the system intended to supply, distribute, and control power and illumination essential for safety to human life."

Emergency power is the most critical category of auxiliary power and is often considered a "life safety" function, as suggested in the definition, since it relates to systems essential for safety of human life. Power for emergency lighting systems falls into this category of auxiliary power.

Legally required power:
"...those systems legally required and classed as emergency by municipal, state, federal or other codes or by governmental agency having jurisdiction. These systems are intended to automatically supply power to selected loads (other than those classed as emergency systems) in the event of failure of the normal source."

Legally required power is typically required to serve loads such as heating and refrigeration, communications systems, ventilation and smoke control systems, sewage disposal, and industrial processes that when stopped could create a hazard or hamper emergency response.

Optional standby power:
"...those systems intended to supply power to public or private facilities or property where life safety does not depend on the performance of the system. These systems are intended to supply on-site generated power to selected loads either automatically or manually."

Optional standby power is typically required to serve loads such as heating and refrigeration, communications systems, ventilation and smoke control systems, sewage disposal, and industrial processes that when stopped would not create a hazard or hamper emergency response.

Critical operations power systems:
"...those systems legally required and classed as emergency by municipal, state, federal or other codes or by governmental agency having jurisdiction or by facility engineering documentation establishing the necessity for such systems. These systems include but are not limited to power systems, HVAC, fire alarm, security, communications, and signaling for designated critical operations."

Control panels, servers and routers often have minimum environmental requirements, so HVAC systems conditioning spaces where equipment is located may also need to be on auxiliary power.

5.2 System Hardening

"Hardened" lighting systems may include some or all of the following:

- Conduit and wiring systems connecting power distribution and control equipment to lighting equipment
- Luminaires, lamps, drivers, ballasts, power supplies
- Central control panels
- Local control devices or field devices for central control equipment
- Seismic restraint, for securing luminaires, panels, and conduits systems

5.2.1 Flooding. FEMA and other authorities produce flood hazard maps defining areas prone to flooding, along with flooding characteristics.[3] Analysis of the flood hazard maps in conjunction with the building siting will reveal a *base flood elevation*. The *design flood elevation* (DFE) is the elevation including wave height. Coastal areas can also be within a special flood hazard area, a *coastal high hazard area* (CHHA), indicating that buildings are subject to high-velocity wave action. In the latter areas, selected walls need to use breakaway construction. Breakaway walls are designed to collapse under wave action, without causing damage to the elevated portion of the building.

5.2.1.1 Protecting Against Floodwaters. The first defense in protecting lighting systems is simply to locate equipment above the DFE, where possible. Communicating this information on drawings, including clearly identifying the DFE, is important in order to avoid unnecessarily routing electrical distribution through areas below the DFE (see **Figure 5-1**). For example, it is common for contractors to route conduit systems in floor slabs. If the floor slab is located below the DFE, conduit will be exposed to floodwaters and may fill with water that will remain trapped in the conduit system.

Where equipment cannot be elevated above the DFE, it is critical that wet floodproofing techniques be employed. For lighting equipment, this requires the use of luminaires and enclosures with submersible ratings, such as IP68 or NEMA 6 (refer to **Annex B**). It is important that conduit systems connecting to dry, flood-proofed equipment also be sealed with conduit seals, such as neoprene or EPDM interior bushings, or with a closed-cell duct sealant. (See **Figure 5-2**.)

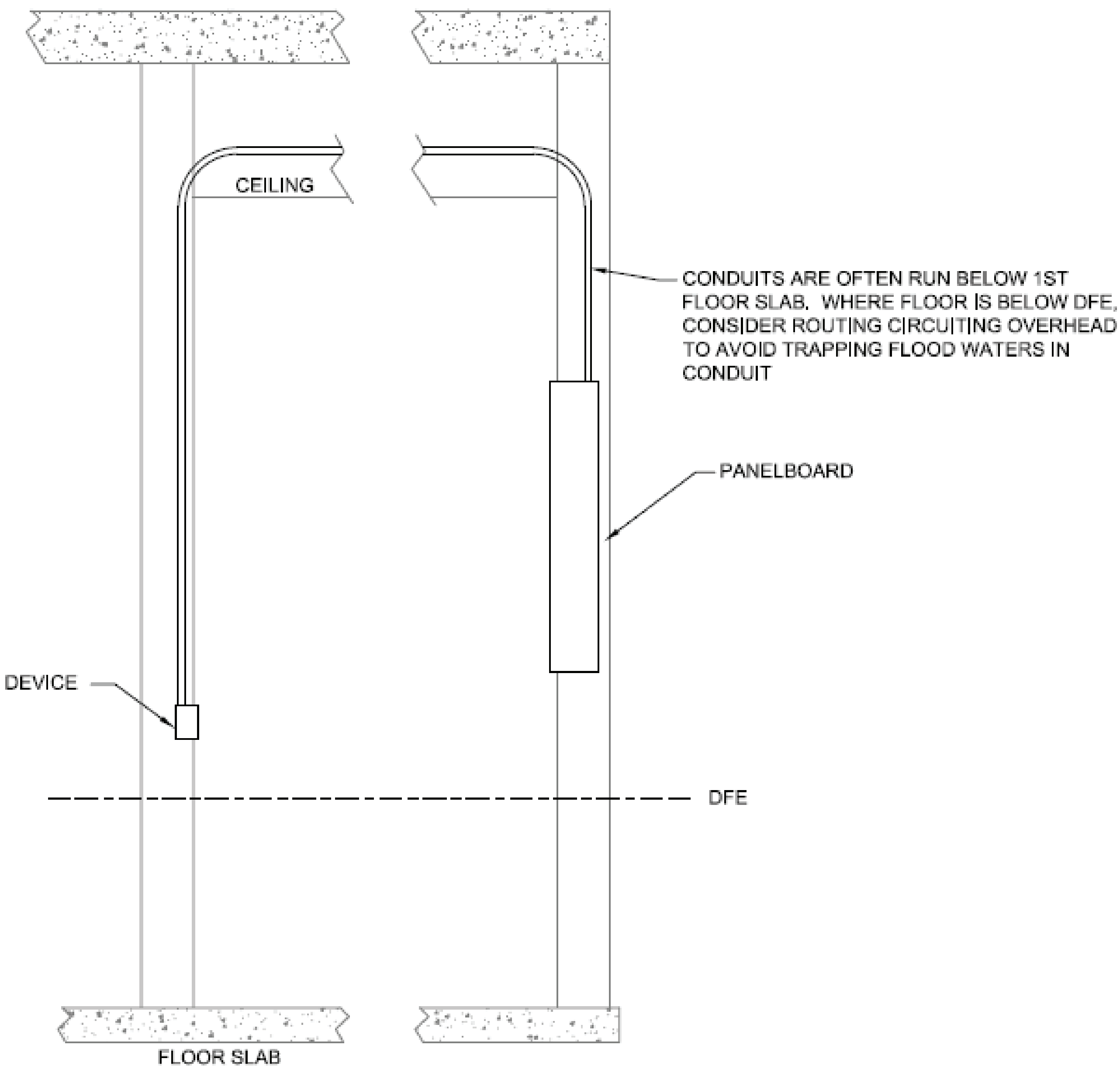

Figure 5-1. Electrical conduits should be routed above the DFE. (Source: Nick Ferzacca.)

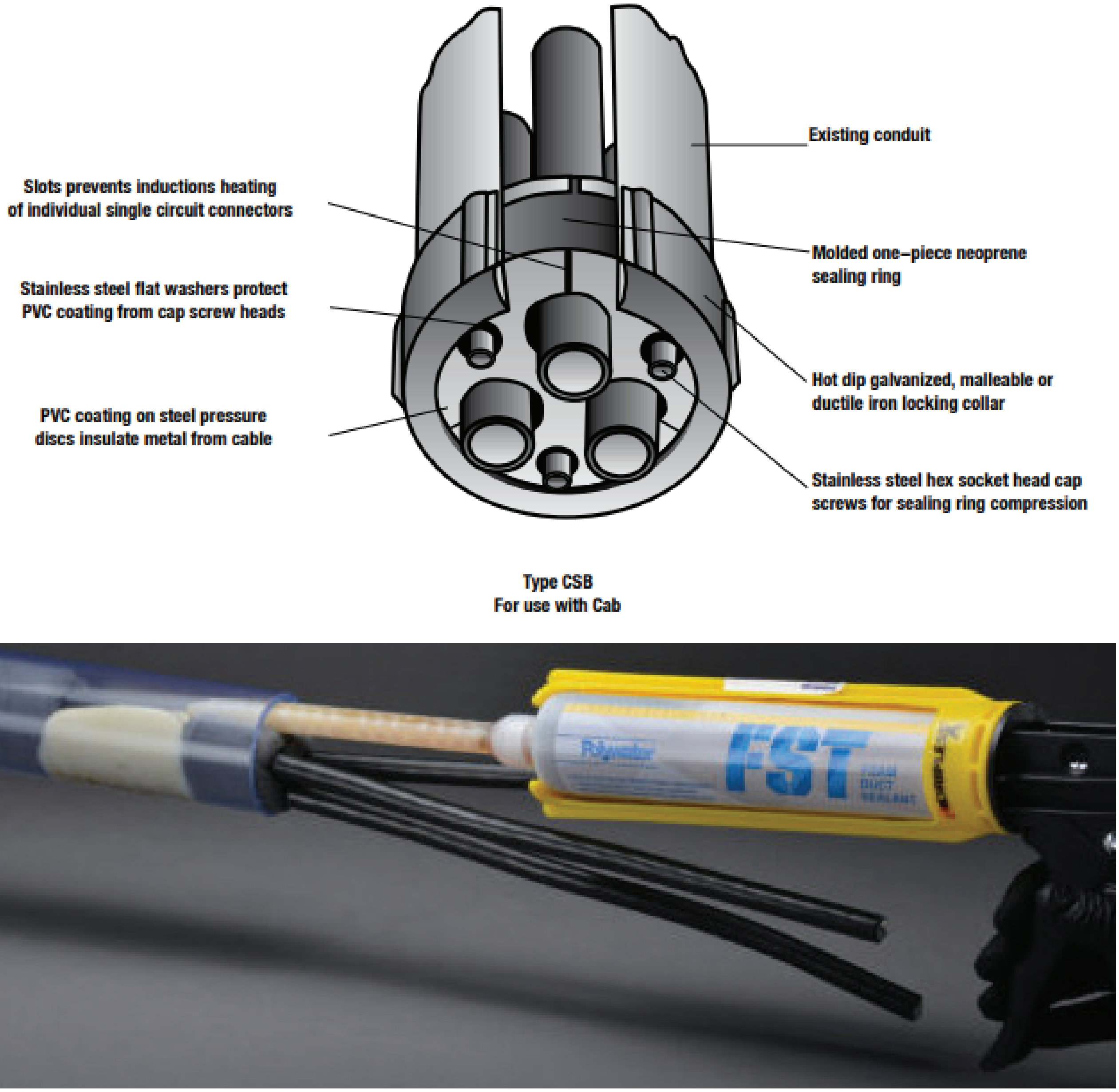

Figure 5-2. *Top:* Example of a sealing bushing; *bottom:* example of a duct sealant. (*Top:* Courtesy of OZ Gedney; *bottom:* Courtesy of American Polywater Corporation)

5.2.1.2 Facilitating Recovery. Submersible-rated luminaires and enclosures are limited, bulky, and expensive. Where entire occupied floors are located below the DFE, it may unreasonable to expect entire lighting systems to be designed for submergibility. Furthermore, while appropriately rated lighting equipment may survive floodwaters, interior building systems such as walls and ceilings may need to be replaced anyway. Therefore, it may be more appropriate to design systems for easy replacement as opposed to sustaining a flood. This could consist of measures that allow easy removal of luminaires, such as whip or cord-and-plug terminations, or utilizing components of luminaires and equipment that can be easily replaced.

5.2.1.3 Breakaway Construction. FEMA guidelines indicate that utilities should not be attached to breakaway walls, so as not to prevent the breakaway design. In areas where the potential for high velocity wave action requires breakaway construction, it is important that construction documents clearly identify breakaway and non-breakaway walls, to allow appropriate layout of controls, luminaires, and associated conduit.

6.0 Resilient Lighting for Homeowners

Resilient lighting in a residential context is lighting that is designed to provide appropriate and reliable illumination as required by those in residential situations in the event of a weather emergency, natural disaster, or other event where people are directed to stay in their homes (to "shelter in place"), even though normal electrical service may not be available or the building may be damaged.

Elements of resilient lighting for residential applications might include:

- Provision for extended emergency power from batteries, renewable sources (e.g., sun, wind), or onsite generation
- A separate power distribution system wired to specific luminaires installed in strategic areas in the home
- Luminaires—including portable luminaires such as table, floor, and task lights—intended for daily use but which include high-capacity rechargeable batteries
- Luminaires designed for recreational vehicles, boats, and camping activities, typically rated for use on low-voltage power systems and perhaps with built-in rechargeable batteries

In a situation such as the COVID-19 global pandemic, resilient lighting in the home may extend to "shelter-in-place" or "work from home" orders to protect public health. In this scenario, resilient lighting for the home extends beyond lighting for traditional "home" tasks. Lighting that is appropriate for work, school, and video calls in the home would also be considered resilient.

7.0 Future Directions – Uses for Resilient Lighting Systems Beyond Illumination

The solid-state nature and predominance of LED lighting continues to lay the groundwork for using lighting systems for applications beyond illumination.

7.1 Smart Street Lighting and Signaling

Smart street lighting is already using cloud-based lighting controls to reduce energy and improve maintenance. Now with the ease of dimming and color-changing LED lighting, it may be possible to use street lighting on a citywide or regional basis to aid in evacuation. Pole mounted lighting along selected roads working in conjunction with traffic signaling can change color, blink, or "chase," in part or as a whole, to clearly identify evacuation routes. Like escalators that reverse direction from the morning to the nighttime commute for pedestrians, lighting can be manipulated to lead and redirect traffic, even reinforcing the opening of converted highway lanes.

7.2 Lighting as a Platform for Communications

Since LEDs are controlled by electronic circuits, radio, TV, and other types of communication signals can be embedded into the emitted light without affecting its visual functions. Such signals can be captured and interpreted by fixed or portable equipment within the range of the light and used for communication, control, and data transfer.

7.3 Radiant Energy for Disinfection

Ultraviolet germicidal irradiation (UVGI) has been used for disinfection of surfaces, water, and air for over 100 years. With the COVID-19 pandemic, interest in the use of UVGI has reemerged as a potential approach to protecting against the spread of viruses. Additional resources on this subject include *IES CR-2-20-V1, Germicidal Ultraviolet – Frequently Asked Questions*[14] and *ASHRAE Epidemic Task Force – Schools and Universities*.[15]

Annex A – Other Resources

There are many resources currently available providing guidance on disaster planning, recovery, and preparedness. Many of these present methods to conduct risk assessments, categorize extreme events, and quantify performance of buildings with respect to events. While these activities are not the primary role of engineers and designers responsible for lighting design, familiarization can only help the programming and design basis of lighting systems. Several helpful publications are listed in the **References** section. Some others include:

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Annex B – NEMA Ratings and IP Equivalency

This Annex is not part of ANSI/IES LP-13-21, Lighting Practice: Introduction to Resilient Lighting Systems. *It is provided for informational purposes only.*

A comparison of NEMA enclosure ratings and IP ratings is provided in **Table B-1.**

Table B-1. Comparison of NEMA and Ingress Protection (IP) Ratings

NEMA Rating	NEMA Definition	IP Rating	IP Definition	
			Solids	Liquids
1	Enclosures constructed for indoor use to provide a degree of protection to personnel against incidental contact with the enclosed equipment and to provide a degree of protection against falling dirt	**IP10**	1 = Protected against solid foreign objects 50 mm in diameter and greater	0 = Not Protected
2	Enclosures constructed for indoor used to provide a degree of protection to personnel against incidental contact with the enclosed equipment, to provide a degree of protection against falling dirt, and to provide a degree of protection against dripping and light splashing of liquids	**IP11**	1 = Protected against solid foreign objects 50 mm in diameter and greater	1 = Protected against vertically falling water drops
3	Enclosures constructed for either indoor or outdoor used to provide a degree of protection to personnel against incidental contact with the enclosed equipment; to provide a degree of protection against falling dirt, rain, sleet, snow, and windblown dust; and that will undamaged by external formation of ice on the enclosure	**IP54**	5 = Protected against dust; limited ingress permitted (no harmful deposits)	4 = Protected against water sprayed from all directions; limited ingress permitted
3R	Enclosures constructed for either indoor or outdoor used to provide a degree of protection to personnel against incidental contact with the enclosed equipment; to provide a degree of protection against falling dirt, rain, sleet, and snow; and that will be undamaged by external formation of ice on the enclosure	**IP14**	1 = Protected against vertically falling water drops	4 = Protected against water sprayed from all directions; limited ingress permitted
3S	Enclosures constructed for either indoor or outdoor use to provide a degree of protection to personnel against incidental contact with the enclosed equipment; to provide a degree of protection against falling dirt, rain, sleet, snow, and windblown dust; and in which the external mechanism(s) remain operable when ice laden.	**IP54**	5 = Protected against dust; limited to ingress (no harmful deposits)	4 = Protected against water sprayed from all directions; limited ingress permitted
4	Enclosures constructed for either indoor or outdoor use to provide a degree of protection to personnel against incidental contact with the enclosed equipment; to provide a degree of protection against falling dirt, rain, sleet, snow, windblown dust, splashing water, and hose-directed water; and that will be undamaged by the external formation of ice on the enclosure	**IP66**	6 = Totally protected against dust	6 = Protected against strong jets of water from all directions; limited ingress permitted
4X	Enclosures constructed for either indoor or outdoor use to provide a degree of protection to personnel against incidental contact with the enclosed equipment; to provide a degree of protection against falling dirt, rain, sleet, snow, windblown dust, splashing water, hose-directed water, and corrosion; and that will be undamaged by the external formation of ice on the enclosure	**IP66**	6 = Totally protected against dust	6 = Protected against strong jets of water from all directions; limited ingress permitted

5	Enclosures constructed for indoor use to provide a degree of protection to personnel against incidental contact with the enclosed equipment; to provide a degree of protection against falling dirt; against settling airborne dust, lint, fibers, and flyings; and to provide a degree of protection against dripping and light splashing of liquids	**IP52**	5 = Protected against dust; limited ingress (no harmful deposits)	2 = Protected against direct sprays of water up to 15° from the vertical
6	Enclosures constructed for either indoor or outdoor use to provide a degree of protection to personnel against incidental contact with the enclosed equipment; to provide a degree of protection against falling dirt; against hose-directed water and the entry of water during occasional temporary submersion at a limited depth; and that will be undamaged by the external formation of ice on the enclosure.	**IP67**	6 = Totally protected against dust	7 = Protected against the effects of temporary immersion of between 15 cm and 1 m; duration of test 30 minutes
6P	Enclosures constructed for either indoor or outdoor use to provide a degree of protection to the personnel against incidental contact with the enclosed equipment; to provide a degree of protection against falling dirt; against hose-directed water and the entry of water during prolonged submersion at a limited depth; and that will be undamaged by the external formation of ice on the enclosure	**IP67**	6 = Totally protected against dust	7 = Protected against the effects of temporary immersion of between 15 cm and 1 m; duration of test 30 minutes
12 and 12K	Enclosures constructed (without knockouts) for indoor use to provide a degree of protection to personnel against incidental contact with the enclosed equipment; to provide a degree of protection against falling dirt; against circulating dust, lint, fibers, and flying; and against dripping and light splashing of liquids	**IP52**	5 = Protected against dust; limited ingress (no harmful deposits)	2 = Protected against direct sprays of water up to 15° from the vertical
13	Enclosures constructed for indoor use to provide a degree of protection to personnel against incidental contact with the enclosed equipment; to provide a degree of protection against falling dirt; against circulating dust, lint, fibers, and flyings; and against the spraying, splashing, and seepage of water, oil, and noncorrosive coolants.	**IP54**	5 = Protected against dust; limited ingress (no harmful deposits)	4 = Protected against water sprayed from all directions; limited ingress permitted

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Process for Change to an ANSI/IES Standard under Continuous Maintenance

This standard is maintained under continuous maintenance procedures, for which IES has an established and documented program for regular publication of addenda or revisions, including procedures for timely, documented, consensus action on requests for change to any part of the standard. Committee consideration will be given to proposed changes by June 30 of any given year for proposed changes received by the IES Director of Standards no later than December 31 of the previous year.

Submittal Format

Proposed changes must be submitted to the IES Director of Standards in the announced published format. However, changes may be accepted in an earlier published format, if the differences are immaterial to the proposed change submittal. If the Director of Standards concludes that a current form must be utilized, the proposer may be given up to 20 additional days to resubmit the proposed changes in the current format.

Specific changes in the text or values are required and must be substantiated. Any change proposals that do not meet these requirements will be returned to the proposer. Supplemental background documents to support changes submitted may be included.

Submission to the Committee Chair

The Director of Standards shall forward proposed changes received on appropriate forms to the committee chair for assigning to committee members (responders) to develop responses to submitters of proposed changes.

Review and Clarification

Responders shall review proposals and should contact the proposer if necessary for clarification.

Response Recommendation

Designated responders shall draft a recommended committee response, including any recommended changes to the standard. The 'responders' recommended responses shall be submitted to the committee chair in electronic form usable by Society Staff, including any recommended change to the standard in response to proposals received.

Options for Committee response are limited to:

- a) Proposed change accepted for public review without modification
- b) Proposed change accepted for public review with modification
- c) Proposed change accepted for further study
- d) Proposed change rejected

The responders shall provide reasons for any recommendation other than option (a) above.

The designated responders shall not recommend option (c) unless the further study can be completed by October 1 of that year, and providing the Committee can then vote for option (a), (b), or (d) no later than November 15 of that year.

Editing

The Committee chair or his or her designee shall edit the draft responses and circulate the edited drafts to the committee for review.

Form for Proposing Change to an ANSI/IES Standard under Continuous Maintenance

NOTE: Use a separate form for each comment. Submit to the Director of Standards, IES, 120 Wall Street, 17th Floor, New York, NY 10005-4001. Email: standards@ies.org. Fax: 212-248-5017.

1. Submitter: ______________________________
   Affiliation: ______________________________
   Address: ______________________________
   City: ______________ State: ________ Zip: ________ Country: ______________
   Telephone: ______________________________
   Fax: ______________________________
   E-mail: ______________________________

I hereby grant the Illuminating Engineering Society (IES) the non-exclusive royalty rights, including non-exclusive rights in copyright, in my proposals. I understand that I acquire no rights in publication of the standard in which my proposals in this, or other analogous, form are used. I hereby attest that I have the authority and am empowered to grant this copyright release.

Submitter's signature: ______________________________ Date: ______________

2. Title of publications and year published ______________________________
3. Clause (section), sub-clause or paragraph number; and page number: ______________________________
4. My proposal (check one):
   [ ] Change to read as follows
   [ ] Delete and substitute as follows
   [ ] Add new text as follows
   [ ] Delete without substitution

Use underscore to show material to be added (<u>added</u>) and strikethrough for material to be deleted (~~deleted~~). Use additional pages if needed.

5. Proposed change:

____
____
____

6. Reason and substantiation:

____
____
____

Select as applicable:
   [ ] Additional pages are attached. Number of additional pages:
   [ ] Attachments or referenced materials cited in this proposal accompany this proposed change.

Please verify that all attachments and references are relevant, current, and clearly labeled to avoid processing and review delays. Please list your attachments here:

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